SUPERSTORE SALES ANALYSIS REPORT

1. INTRODUCTION

This project analyzes Superstore sales data using Python. The goal is to find insights about sales, profit, regions, and product performance.

2. DATASET OVERVIEW

The dataset contains:
- Sales information
- Profit values
- Product categories
- Customer segments
- Regions
- Order dates

3. DATA CLEANING

- Removed duplicate values
- Handled missing values
- Converted date columns to datetime format
- Fixed column names

4. FEATURE ENGINEERING

New columns created:
- Year
- Month
- Month Name
- Day

5. KEY PERFORMANCE INDICATORS (KPIs)

- Total Sales = (add value here)
- Total Profit = (add value here)
- Profit Margin = (add value here)

6. REGIONAL ANALYSIS

- West region performed best in sales
- Some regions have lower performance

7. CATEGORY ANALYSIS

- Technology category is most profitable
- Furniture category has lower profit

8. TOP PRODUCTS

- Top 10 products identified based on sales
- These products contribute most to revenue

9. LOSS MAKING PRODUCTS

- Some products generate negative profit
- These should be improved or removed

10. MONTHLY SALES TREND

- Sales change across months
- Some months show peak performance

11. KEY INSIGHTS

- Technology is most profitable category
- West region is strongest
- Some products are causing losses
- Sales show seasonal patterns

12. CONCLUSION

This analysis helps understand business performance and provides insights to improve sales and profit.

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```
19 month
20 month_name
21 day
```

[76]:

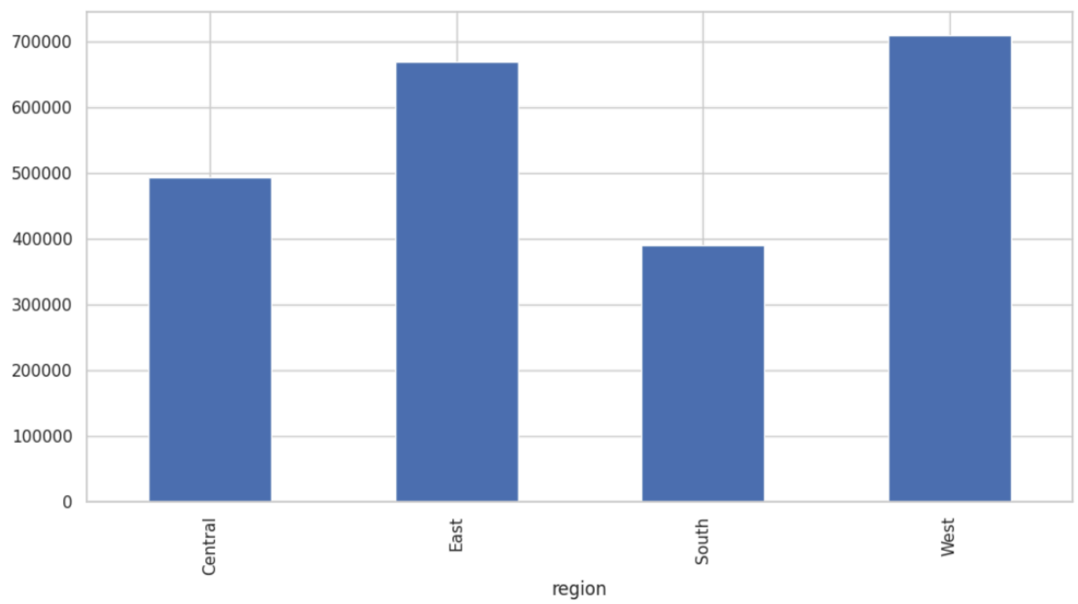
```
df.head()
```

[76]:

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	city	...	region	produ
0	1	CA-2017-152156	2017-08-11	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	South	FUR 1000
1	2	CA-2017-152156	2017-08-11	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	South	FUR 1000
2	3	CA-2017-138688	2017-12-06	NaT	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	West	OFF 1000
3	4	US-2016-108966	2016-11-10	NaT	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	...	South	FUR 1000

US-

```
print("Region column not found, skipping region analysis")
```



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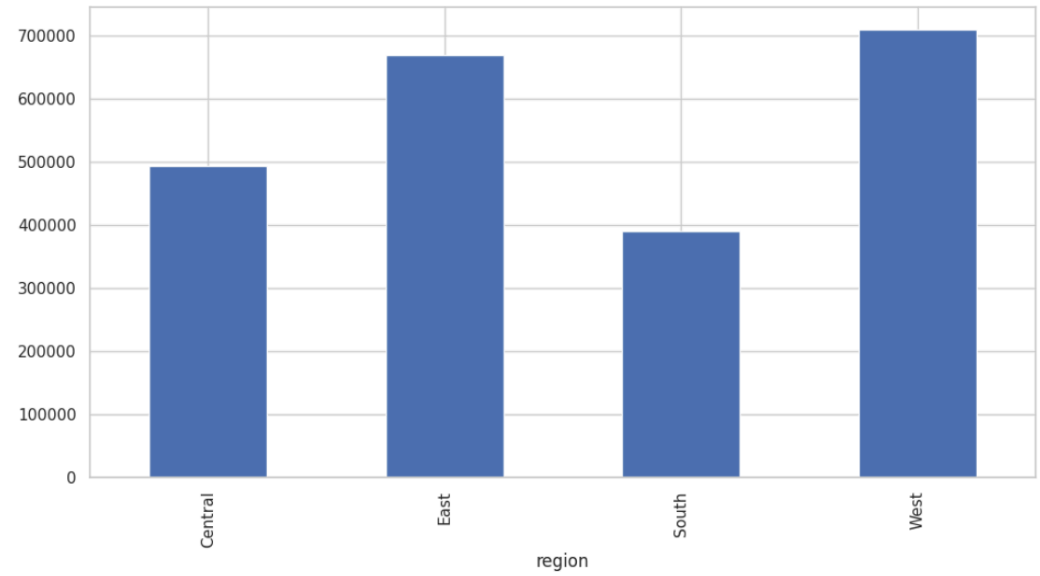
```
else:
    print("Region column not available")
```



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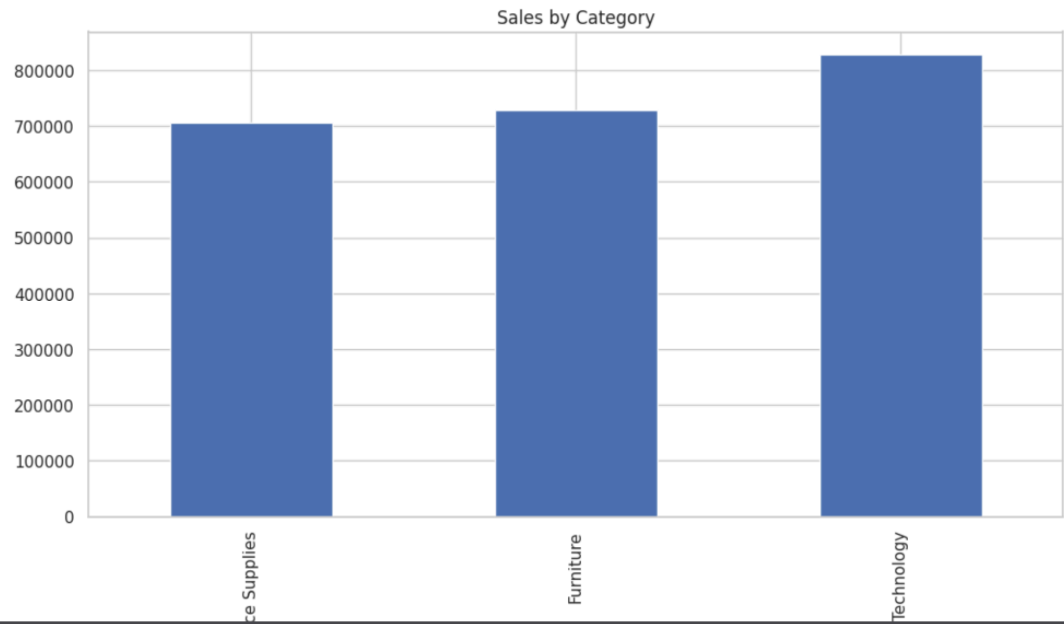
```
    print("Category column not found")
```



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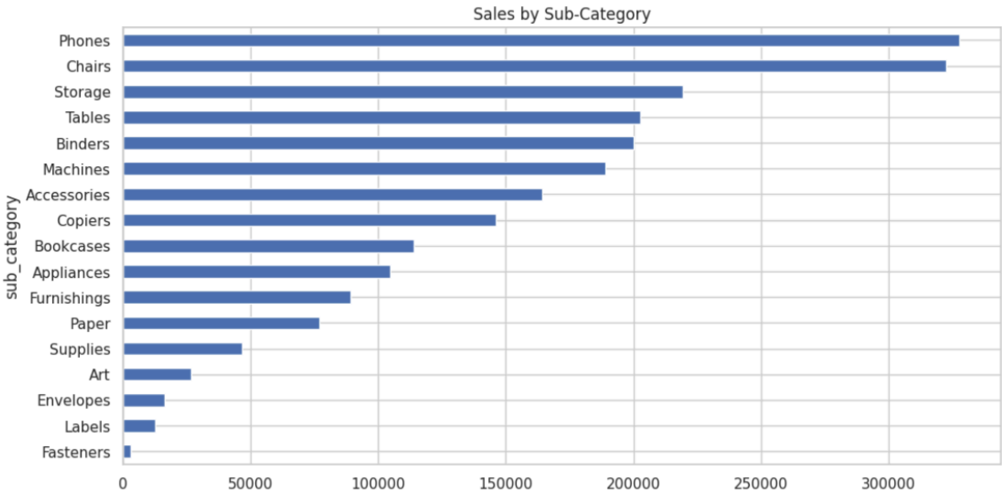
```python
    import matplotlib.pyplot as plt
    plt.title("Sales by Sub-Category")
    plt.show()
else:
    print("Sub-category column not found")
```



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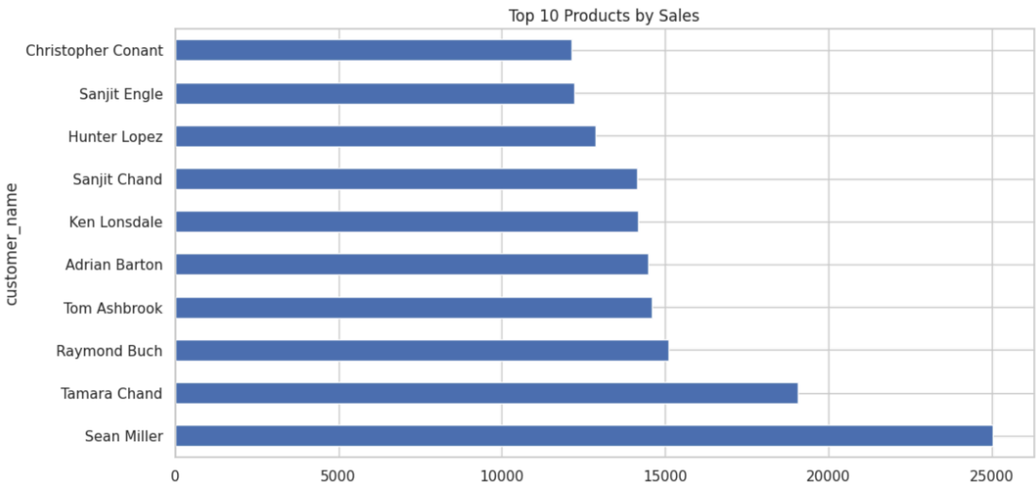
```python
    df.groupby(product_cols[0])[sales_cols[0]].sum().sort_values(ascending=False).head(10).plot(kind= ba
    import matplotlib.pyplot as plt
    plt.title("Top 10 Products by Sales")
    plt.show()
else:
    print("Product column not found")
```



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```
    plt.show()
else:
    print("Monthly data not available")
```